

AI-Driven Multi-Brand Social Media Content Management System

Project Proposal



Supervisor
Nomaan Khan

Submitted by
Shahid Yasin
BSCS-E1-22-03

Department of Computer Science,
University of Sahiwal.

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Project Title:

AI-Driven Multi-Brand Social Media Content Management System

Submitted by:

Student Name: Shahid Yasin

Roll No: BSCS-E1-22-03

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Supervisor:

Name: Nomaan Khan

Signatures:

Chairman Approval:

Name: Dr. M Ayyoub

Signature:

Date:

Comments (if any):

Department Seal:

1. Project Title

AI Driven Multi-Brand Social Media Content Management System

2. Introduction

With the rapid expansion of digital marketing, organizations increasingly rely on structured systems to manage and automate their social media strategies. This project proposes a backend-centric web platform designed to manage content workflows for multiple brands operating within the same niche. The system models brand identities, processes structured trend datasets, generates content drafts, applies validation rules, and manages approval and scheduling mechanisms. The focus is on system design, automation pipelines, multi-tenant database modeling, and asynchronous task management.

3. Objective

To design and implement a scalable backend system that manages multi-brand content workflows through structured data processing, rule-based validation, scheduling mechanisms, and database-driven automation.

4. Problem Description

Brands operating within similar niches often compete for the same audience segments. Managing consistent content strategies while maintaining distinct brand identity becomes challenging when workflows are handled manually or through disconnected tools.

Existing solutions primarily focus on either content generation or scheduling, lacking a unified system that integrates persona modeling, validation mechanisms, structured approval workflows, and timed execution within a centralized backend architecture.

Furthermore, when multiple brands are managed under a single system, issues such as data overlap, inconsistent brand tone, and inefficient workflow coordination may arise. There is a need for a structured, rule-driven system capable of handling brand separation, workflow automation, and content lifecycle management.

The proposed solution addresses these issues by designing a modular system that separates data ingestion, persona configuration, validation logic, scheduling services, and storage layers. The emphasis is on backend engineering principles including database relationships, automation pipelines, asynchronous processing, and system scalability.

5. Methodology

The system will follow a modular layered architecture:

Backend Development:

FastAPI will be used to implement asynchronous RESTful APIs. The system will

include modules for brand management, content lifecycle handling, validation services, and scheduling operations.

Database Design:

PostgreSQL will be used to implement a relational database schema supporting multi-brand data isolation. SQLAlchemy ORM will manage model relationships and transactions.

Automation & Scheduling:

APScheduler will be integrated for background task execution such as scheduled publishing and workflow triggers.

Validation Engine:

A rule-based filtering mechanism will enforce brand persona parameters and content validation constraints.

System Architecture:

The application will follow a layered structure consisting of:

- API Layer
- Service Layer
- Data Access Layer
- Scheduler Module

Testing will include unit testing, integration testing, and API validation using Postman.

6. Project Scope

Included:

- Multi-brand persona modeling
- Structured content workflow management
- Approval and rejection mechanisms
- Database-driven scheduling simulation
- REST-based backend architecture
- Web-based administrative dashboard (React.js)

Not Included:

- Real-time integration with social media platforms
- Payment systems
- Advanced AI model training
- Mobile application development
- Distributed microservices deployment

Assumptions:

- Structured trend datasets will be available for processing.
- System will operate in a centralized backend environment.
- Users manage brand configurations independently.

7. Feasibility Study

i. Risks Involved

Scope Expansion:

Controlled by strict modular planning and milestone tracking.

Complex Database Relationships:

Mitigated through proper ER modeling before implementation.

Scheduling Conflicts:

Handled through structured time-based validation checks.

ii. Resource Requirements

- Development PC (minimum 8GB RAM)
- PostgreSQL database server
- Python development environment
- Git version control
- No specialized hardware required

The project is technically feasible within the academic timeline.

8. Solution Application Areas

Target Industries:

- Digital Marketing Agencies
- Brand Management Firms
- Content Strategy Departments
- Social Media Operations Teams

Benefits:

- Structured multi-brand workflow management
- Reduced manual coordination effort
- Clear separation of brand identities
- Automated lifecycle control
- Scalable backend architecture

The system provides practical value in improving workflow efficiency and structured content operations.

9. Tools/Technology

- **Python** – Backend programming language
- **FastAPI** – Backend framework for REST APIs
- **PostgreSQL** – Relational database
- **SQLAlchemy** – ORM for database interaction
- **APScheduler** – Background task scheduling

- **React.js** – Frontend dashboard interface
- **Git** – Version control
- **Postman** – API testing tool

10. Expertise of the Team Members

The developer possesses foundational knowledge in:

- Object-Oriented Programming
- Database Systems
- Data Structures
- Web Development
- REST API Design

Practical experience with Python and backend development ensures the capability to implement scalable API-driven systems.

The project aligns with prior coursework in Software Engineering and Database Management Systems. Additional research will focus on workflow automation and multi-tenant backend design.

11. Milestones

Milestone 1: Requirement Analysis & System Design

- Finalize scope
- Design ER diagram
- Define API structure

Milestone 2: Database & Model Implementation

- Implement PostgreSQL schema
- Define ORM models

Milestone 3: Backend API Development

- Brand management APIs
- Content lifecycle APIs
- Validation module

Milestone 4: Scheduler Integration

- Implement automated scheduling
- Background job testing

Milestone 5: Dashboard Development

- Build admin interface
- Integrate APIs

Milestone 6: Testing & Optimization

- Unit testing
- Integration testing
- Performance validation

Milestone 7: Documentation & Final Deployment

- Complete report
- Prepare presentation
- System demonstration

12. References

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